

Chromatography

Answer key with worked solutions

A. Understanding

1. Phases.

(a) The **stationary phase** stays in place (paper, silica, a column coating); the **mobile phase** moves through or over it (a solvent or a gas). (b) Each substance is shared between the two phases according to its **attraction** to each. A substance held more strongly by the stationary phase spends more time there and **moves more slowly**, so the substances travel different distances and separate.

B. R_f values

2. R_f values.

$$(a) \frac{2.10 \text{ cm}}{8.40 \text{ cm}} = 0.250; \frac{4.62 \text{ cm}}{8.40 \text{ cm}} = 0.550; \frac{6.72 \text{ cm}}{8.40 \text{ cm}} = 0.800$$

(b) **X** (0.25) and **Y** (0.55). (c) The spot at R_f 0.800 matches **none** of the standards (Z is 0.70): it is a different, unidentified substance.

3. Technique.

(a) Pencil (graphite) does not dissolve; **ink would separate** and add spots. (b) So the spots are **not washed into the solvent**. (c) To keep the air **saturated with solvent vapour**, so the solvent does not evaporate from the plate and the front moves evenly. (d) The front **evaporates quickly** and becomes invisible, and its distance is needed for R_f.

C. Polarity

4. Polarity.

(a) **Hexane > propanone > ethanoic acid**. Silica is very polar. Non-polar hexane is held least and moves furthest; propanone is polar (C=O); ethanoic acid is the most polar and can **hydrogen-bond** to the silica, so it moves least. (b) A more polar solvent competes better for the compounds, so **all the R_f values increase**.

5. One spot.

No. Two substances can have the **same R_f** in one solvent. Run the sample again with a **different solvent** (or stationary phase), or use another method such as GC or a melting point; a pure sample gives one spot under every condition.

D. Gas chromatography

6. Retention times.

(a) **Ethanol** and **butan-1-ol** (the retention times match the standards; no propan-1-ol).
(b) Ethanol has the **lowest boiling point** (smallest molecule, weakest attractions), so it spends most time in the gas phase and leaves the column first.

7. Area percent.

$$\text{Total} = 2310 + 1540 + 3850 = 7700$$

$$\frac{2310}{7700} \times 100 \% = 30.0 \% ; \frac{1540}{7700} \times 100 \% = 20.0 \% ; \frac{3850}{7700} \times 100 \% = 50.0 \%$$

E. Replicates and quantities

8. Quintuplicate R_f.

(a) Mean = $\frac{0.61 + 0.63 + 0.62 + 0.60 + 0.64}{5} = 0.62$; $s = 0.016$ (sample standard deviation, $n - 1 = 4$)

(b) **Dye M.** Its R_f (0.62) equals the mean. Dye N (0.71) differs from the mean by 0.09, more than 5 standard deviations, so it does not match.

9. GC quantification.

(a) Mean area = $\frac{1850 + 1870 + 1860}{3} = 1860$; $s = 10$; RSD = $\frac{10}{1860} \times 100 \% = 0.54 \%$

(b) $c = 0.500 \text{ mg/mL} \times \frac{1860}{2480} = 0.375 \text{ mg/mL}$ (the areas cancel)

(c) $m = 0.375 \text{ mg/mL} \times 20.0 \text{ mL} = 7.50 \text{ mg}$